APPLIED MATHEMATICS - UBA0620

Innovative Assignment - Slot C

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Course Code & Name	UBA0620 / Applied Mathematics
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Academic Year / Batch	2026
Date of Submission	03.09.2026
Maximum Marks	100

1. Problem Understanding and System Overview

The engineering scenario at the centre of this assignment is a **specialty-chemicals plant** where an exothermic batch reactor must be held within a strict temperature window throughout its reaction cycle. Coolant flows through a jacket surrounding the reactor vessel, absorbing heat generated by the reaction. The challenge is mathematically rich: prevent the temperature from climbing too high (thermal runaway risk) while also avoiding over-cooling (which slows conversion and wastes energy).

Five branches of applied mathematics are brought together to analyse one real system. Each tool addresses a different question about the same reactor:

- **How does the coolant move, and how much exits through the jacket wall?** -- Vector Calculus
- **How can the repeating start-up/hold/shutdown temperature profile be represented?** -- Fourier Series
- **How does reactor temperature respond when the coolant valve is suddenly upset?** -- Laplace Transform
- **What frequency components does the noisy flow sensor pick up?** -- Fourier Transform & Parseval's Identity
- **Is the discrete-time PID controller that governs the valve digitally stable?** -- Z-Transform

1.1 Selected Parameters

The table below lists specific values adopted for every calculation. Where the problem statement allowed freedom, physically realistic values were selected.

Quantity	Symbol	Value / Definition
Coolant velocity field	F(x,y,z)	(2x-y)i + (yz^2)j + (y^2*z)k m/s
Jacket surface (closed)	S	Cylinder: r = 0.5 m, h = 1.2 m
Inlet-baffle surface	Sigma	Square patch 0 <= x,y <= 0.5 m at z = 0
Initial reactor temperature	T0	30 degrees C
Heat-transfer constant	k	0.5 per second
Valve-upset disturbance	f(t)	10 u(t) (unit step of amplitude 10)
Sensor signal	v(t)	e^(-0.5t) sin(2*pi*t), t >= 0
Sampling interval	Ts	0.1 s
PID gains	Kp, Ki, Kd	1.0, 0.2, 0.02

Table 1 - Numerical parameters used throughout the assignment

1.2 Key Assumptions

1. The velocity field F is smooth and continuously differentiable throughout the jacket volume -- a requirement for both Stokes' and the Divergence theorem.

2. The cooling jacket is idealised as a perfect closed cylinder; end-cap thickness and pipe connections are neglected.

3. Reactor thermal dynamics are captured by a first-order linear ODE -- a valid approximation when wall thermal resistance dominates.

4. The sensor signal is modelled as a deterministic decaying sinusoid; stochastic noise is not carried through analytical derivations.

5. The digital PID operates at a fixed, known sampling rate with no computation delay.

2. Vector Calculus -- Coolant Circulation and Jacket Flux

2.1 Circulation via Stokes' Theorem

Stokes' theorem converts a line integral around a closed curve C into a surface integral of the curl over any surface bounded by C:

```
Loop integral over C of F dot dr  =  Double integral over Sigma of
                                     (curl F) dot n dS
```

Computing the curl component-by-component from F = (2x-y)i + (yz^2)j + (y^2*z)k :

- i-component: d(y^2*z)/dy - d(yz^2)/dz = 2yz - 2yz = 0
- j-component: d(2x-y)/dz - d(y^2*z)/dx = 0 - 0 = 0
- k-component: d(yz^2)/dx - d(2x-y)/dy = 0 - (-1) = 1

Therefore: curl F = k (a uniform unit vector in the z-direction)

The baffle surface is the 0.5 m x 0.5 m square in the z = 0 plane, with outward normal n = k:

```
Circulation  =  Double integral over Sigma of  (k dot k) dA  =
                Area  =  (0.5)(0.5)  =  0.25 m^2/s
```

Key Points: Physical Interpretation

- A circulation of +0.25 confirms a net counter-clockwise swirl entering the jacket at the baffle.
- Because curl F is uniform (= k), the circulation equals simply the area of the chosen surface.
- A non-zero swirl is actually desirable -- it enhances heat-transfer mixing inside the jacket.

2.2 Net Outward Flux via the Divergence Theorem

For a closed surface S, the Divergence theorem relates the outward flux to the volume integral of the divergence:

```
Closed surface integral of F dot n dS  =  Triple integral over V of
                                          (div F) dV
```

The divergence of the field:

```
div F  =  d(2x-y)/dx  +  d(yz^2)/dy  +  d(y^2*z)/dz  =  2  +  z^2
                                                        +  y^2
```

Converting to cylindrical coordinates and integrating over the cylinder (r=0.5 m, h=1.2 m):

- Constant term: 2 x pi x r^2 x h = 2 x pi x 0.25 x 1.2 = 1.885
- z^2 term: pi x r^2 x (h^3/3) = pi x 0.25 x 0.576 = 0.452
- r^2*sin^2(theta) term: (pi/4)(r^4/2) x h = (pi/4)(0.0313/2)(1.2) = 0.059

```
Net outward flux  =  1.885 + 0.452 + 0.059  =  approximately 2.396
                                               m^3/s
```

Key Points: Physical Interpretation

- A positive flux indicates a net outward flow tendency across the entire jacket surface.
- In a real system this would signal net coolant generation inside the jacket -- meaning the idealised field is a flow-balance reference rather than an exact measurement.
- The practical value: if a measured flux differs significantly from this model value, it flags a sensor error or coolant leak.

3. Fourier Series -- Periodic Temperature Profile

3.1 Choosing the Waveform

During a typical batch reaction the temperature follows a recognisable pattern: it ramps up quickly during start-up, holds roughly constant during the main reaction, then drops during shutdown. A square wave of period 2*pi is a compact mathematical model of this alternating behaviour.

The temperature deviation from the set point is modelled as:

```
T_d(t)  =  10 x sign(sin t)        [period = 2*pi]
```

This function is odd (T_d(-t) = -T_d(t)), which immediately tells us:

- The Fourier series contains only sine terms -- no cosine terms and no constant term.
- This halves the computation compared to a general periodic function.

3.2 Fourier Coefficients

For an odd function, the only non-zero coefficients are b_n:

```
b_n  =  (1/pi) Integral from 0 to 2*pi of T_d(t) sin(n*t) dt
   b_n  =  40/(n*pi)   for odd n;    b_n = 0   for even n
```

The Fourier series is therefore:

```
T_d(t)  =  (40/pi) [ sin(t)  +  sin(3t)/3  +  sin(5t)/5  +
                     sin(7t)/7  + ... ]
```

3.3 Harmonic Contributions and Convergence

Harmonic (n)	Coefficient bn = 40/(n*pi)	Contribution at t = pi/2
1	12.732	+12.732
3	4.244	-4.244
5	2.546	+2.546
7	1.819	-1.819
9	1.415	+1.415
Sum (5 harmonics)	--	approx. 10.63 (exact = 10)

Table 2 - Fourier coefficients for the square-wave temperature model

At a jump discontinuity, the Fourier series converges to the average of the left and right limits (zero at t = pi). Close to the jump, the partial sum overshoots by roughly 9% -- this is the **Gibbs phenomenon**. It is a mathematical property of the series, not a modelling error, and the overshoot region narrows as more harmonics are added.

4. Laplace Transform -- Response to a Coolant-Valve Upset

4.1 The Governing Equation

A first-order energy balance for the reactor temperature:

```
dT/dt  +  k*T  =  f(t),          T(0) = T0
```

Selected values: k = 0.5 per second, T0 = 30 degrees C, f(t) = 10*u(t) (step disturbance).

4.2 Solution in the Laplace Domain

Applying the Laplace transform and using the initial condition:

```
s*T(s)  -  30  +  0.5*T(s)  =  10/s
T(s) * (s + 0.5)  =  30  +  10/s  =  (30s + 10) / s
T(s)  =  (30s + 10) / [ s * (s + 0.5) ]
```

Partial-fraction decomposition: T(s) = A/s + B/(s + 0.5)

Finding A (multiply both sides by s, set s = 0):

```
10  =  A * 0.5      -->      A = 20
```

Finding B (compare coefficients of s):

```
A + B = 30      -->      B = 10
```

Therefore:

```
T(s)  =  20/s  +  10/(s + 0.5)
```

4.3 Time-Domain Response and Verification

```
T(t)  =  20  +  10 * e^(-0.5t)          (t >= 0)
```

Check	Calculation	Result
Initial condition T(0)	20 + 10 * e^0 = 20 + 10	30 degrees C (matches T0 exactly)
Steady-state T(infinity)	20 + 10 * 0 = 20	20 degrees C
2% settling time	4/k = 4/0.5	8 seconds
Stability check	k = 0.5 > 0	Stable -- exponential decays to zero

Table 3 - Verification of the Laplace-domain solution

Key Points: Engineering Insight

- The reactor temperature drops from 30 deg C toward a new steady state of 20 deg C after the valve upset -- a 10 deg C change settling in approximately 8 seconds.
- A larger k (tighter heat exchange) shortens this settling time; a smaller k means the temperature takes longer to stabilise.
- The positive value of k guarantees exponential decay -- the reactor will never diverge under this linear model.

5. Fourier Transform -- Frequency Analysis of the Flow Sensor

5.1 Sensor Signal Model

The coolant-flow sensor returns a signal that decays after a step change in flow rate and oscillates at a characteristic frequency. This is modelled as:

```
v(t)  =  e^(-0.5t) * sin(2*pi*t),        t >= 0
```

Key features:

- Decay rate alpha = 0.5 per second -- the envelope halves approximately every 1.4 seconds.
- Oscillation frequency f0 = 1 Hz (omega_0 = 2*pi rad/s) -- the dominant spectral line expected in the output.

5.2 Analytical Fourier Transform

Using the standard transform pair for a one-sided decaying sinusoid:

```
Fourier{e^(-alpha*t) sin(omega_0 * t) u(t)}  =  omega_0 / [ (alpha
                    + j*omega)^2  +  omega_0^2 ]
```

Substituting alpha = 0.5, omega_0 = 2*pi:

```
V(omega)  =  2*pi / [ (0.5 + j*omega)^2  +  4*pi^2 ]
```

The magnitude |V(omega)| peaks at omega = 2*pi, confirming that signal energy is concentrated near 1 Hz.

5.3 Energy Check Using Parseval's Identity

Parseval's identity states that total signal energy is the same whether computed in the time domain or the frequency domain:

```
Integral of |v(t)|^2 dt  =  (1/2*pi) * Integral of |V(omega)|^2
                            d_omega
```

Time-domain energy (analytical):

```
E  =  Integral from 0 to infinity of e^(-t) * sin^2(2*pi*t) dt  =
                    approximately 0.497 J
```

The Python FFT-based numerical estimate gives approximately 0.497 J (within 0.1% over a 20-second window), confirming consistency between time- and frequency-domain descriptions.

Key Points: What This Means for the Cooling System

- Frequency analysis lets engineers separate a genuine 1 Hz flow oscillation from higher-frequency electronic noise.
- A notch filter centred on 1 Hz could remove this oscillation if it were an artefact rather than a real flow pattern.
- Parseval's identity provides a quick sanity check: if time-domain energy and spectral energy disagree, there is a numerical error in the FFT or sampling rate.

6. Z-Transform -- Stability of the Digital PID Controller

6.1 Discrete-Time System Setup

The coolant valve is governed by a digital PID controller that reads the temperature error once every Ts = 0.1 s and outputs a valve-position command. The discrete plant is approximated by:

```
G(z)  =  0.2 / (1  -  0.8 * z^(-1))
```

The standard discrete-PID pulse transfer function is:

```
C(z)  =  Kp  +  Ki*Ts / (1 - z^(-1))  +  Kd*(1 - z^(-1)) / Ts
```

With the chosen gains Kp = 1.0, Ki = 0.2, Kd = 0.02:

```
C(z)  =  1.0  +  0.02/(1 - z^(-1))  +  0.2*(1 - z^(-1))
```

6.2 Closed-Loop Poles and Stability

The closed-loop transfer function is:

```
T_cl(z)  =  C(z)*G(z) / [ 1 + C(z)*G(z) ]
```

Computing the combined numerator and denominator polynomials and finding the roots gives the closed-loop poles:

Pole	Location (z-plane)	|z|	Stability Status
z1	approximately 0.990	0.990	Inside unit circle -- STABLE
z2	approximately 0.682	0.682	Inside unit circle -- STABLE

Table 4 - Closed-loop pole locations and stability status

Since both poles lie strictly inside the unit circle (|z| < 1), the discrete closed-loop system is asymptotically stable for the selected tuning.

Key Points: Key Points on Discrete-Time Stability

- In continuous-time (Laplace domain), stability requires poles in the left half-plane.
- In discrete-time (Z-domain), the equivalent condition is poles strictly inside the unit circle (|z| < 1).
- The pole near z = 0.990 indicates a slow mode -- almost on the unit circle -- suggesting integral gain Ki could be slightly increased to speed response without losing stability.
- The pole at z = 0.682 corresponds to the faster derivative-dominated mode, which is well inside the circle and poses no stability concern.

7. Python Verification Strategy

Python was used exclusively to verify results already derived analytically -- not to replace mathematical reasoning. The table below maps each analytical result to its computational check:

Analysis Part	Analytical Result	Python Verification Method
Stokes' theorem	Circulation = 0.25	Numerical surface integral of curl F over Sigma
Divergence theorem	Net flux approx. 2.396	Numerical triple integral over cylinder volume V
Fourier series	Odd-harmonic series	Plot of partial sums (n=1,3,5,7,9) vs exact square wave
Laplace transform	T(t) = 20 + 10*e^(-0.5t)	solve_ivp on original ODE; overlaid with analytical curve
Fourier transform	Peak at 1 Hz	np.fft of sampled v(t); peak identified at 1 Hz index
Parseval's identity	E approx. 0.497 J	np.trapz on |v|^2 vs (1/2*pi)*sum(|V|^2)*delta_omega
Z-transform poles	|z1,z2| < 1	np.roots on closed-loop denominator polynomial

Table 5 - Mapping between analytical derivations and Python verification steps

The Python output in every case agreed with the hand-derived result to within numerical precision (finite time-window and discretisation error), confirming that neither the algebra nor the code contains a systematic mistake.

8. Consolidated Results and Engineering Decision

8.1 Results at a Glance

Part	Method	Key Result	Engineering Meaning
(a)	Stokes' theorem	Circulation = 0.25 m^2/s	Confirmed net inlet swirl -- promotes mixing
(a)	Divergence theorem	Flux approx. 2.396 m^3/s	Flow-balance reference for jacket model
(b)	Fourier series	Odd harmonics; 5 terms approx. 90% accurate	Compact periodic temperature representation
(c)	Laplace transform	T(t) = 20 + 10*e^(-0.5t); settles in 8 s	Stable reactor; cools 10 deg from T0 after upset
(d)	Fourier transform	Dominant frequency = 1 Hz	Flow oscillation identifiable; filterable if needed
(d)	Parseval's identity	E_time approx. E_freq approx. 0.497 J	Spectral description is energy-consistent
(e)	Z-transform	Poles at 0.990 and 0.682	Digital PID is stable for chosen gains

Table 6 - Full results summary

8.2 Safety and Performance Recommendation

1. Thermal stability: The first-order model with k = 0.5 per second predicts a settling time of 8 seconds after a valve upset. This is sufficiently fast for a specialty-chemicals process where reaction rates typically change on a time scale of minutes.

2. Controller stability: Both closed-loop poles of the digital PID lie inside the unit circle -- the sampled controller is confirmed stable at Ts = 0.1 s. The pole close to z = 1 suggests integral action could be slightly increased to reduce steady-state error without losing stability.

3. Sensor performance: The 1 Hz oscillation in the sensor signal is clearly identifiable in the spectrum. If traced to pump vibration, a notch filter would remove it cleanly. Parseval's identity confirms total signal energy is preserved in the frequency domain.

4. Model limitations: The mathematical model is idealised. A real reactor will have nonlinear reaction kinetics, time-varying heat-transfer coefficients, and coloured noise. The linear first-order model is valid only for small deviations around the operating point.

9. Broader Considerations

9.1 Safety

Exothermic batch reactors represent a genuine industrial hazard. The mathematical model developed here can act as an early-warning tool: if the computed settling time extends beyond a safe threshold, process engineers know that the cooling capacity is insufficient before the reactor is built or operated.

9.2 Sustainability Linkages

SDG	Connection to this Work
SDG 3 -- Good Health & Well-being	Preventing thermal runaway protects plant workers and surrounding communities from injury.
SDG 7 -- Affordable & Clean Energy	A well-tuned digital controller minimises unnecessary coolant pumping, reducing utility energy consumption.
SDG 9 -- Industry, Innovation & Infrastructure	The mathematical toolkit from vector calculus to Z-transforms is directly used by process-safety and control engineers in industry.
SDG 12 -- Responsible Consumption	Optimised cooling reduces batch losses and excess coolant usage, improving resource efficiency.

Table 7 - SDG mapping for this engineering scenario

10. Student Reflection

When I first read the problem statement, it looked like five separate mathematics problems bolted together. By the time I finished, I realised they are really five different lenses on the same system.

The part that surprised me most was how directly the Z-transform stability criterion (poles inside the unit circle) mirrors the Laplace stability criterion (poles in the left half-plane). Once I understood why the unit circle in the z-plane corresponds to the imaginary axis in the s-plane, the digital control analysis felt intuitive rather than mechanical.

Python was useful not just as a calculation tool, but as an honesty check. When my hand-computed Parseval energy (approximately 0.497 J) matched the numerical FFT value to three decimal places, it gave me genuine confidence in the derivation rather than just a number on a page.

If I were to extend this work, I would replace the idealised first-order temperature model with a distributed-parameter model that accounts for axial temperature gradients along the jacket height -- a partial-differential-equation problem that connects directly to heat-equation topics in the next semester.

References

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